Ajanta: Echoes in Ochre — The Living Murals of Ajanta

An Interactive Digital Storytelling Website Inspired by the Ajanta Cave Paintings

Course Project: Interactive Media & Web Engineering | Live Website: https://ajanta-echoes-in-ochre.vercel.app/

1. Project Overview

Ajanta: Echoes in Ochre is a responsive, single-page digital museum and narrative storytelling website developed as an academic project to explore the artistic, cultural, and moral heritage of the UNESCO World Heritage Ajanta Caves in Maharashtra, India. Dating from approximately the 2nd century BCE to the 5th–6th century CE, the rock-cut monuments at Ajanta represent the pinnacle of ancient Indian painting and Buddhist rock architecture. However, due to geographical distance and environmental fragility, these masterworks remain largely inaccessible to modern students and general audiences in their original narrative context.

The primary objective of this project was to translate the ancient mural storytelling tradition into an engaging, interactive digital experience while preserving its historical dignity. Rather than constructing a detached image gallery or an oversimplified slideshow, the website integrates verified high-resolution photographs of surviving Cave 17 murals with an illustrated, five-scene adaptation of a traditional Buddhist moral tale. By utilizing thoughtful typography, an authentic natural mineral pigment palette, and scroll-driven animation, the website recreates the contemplative experience of walking through the dimly lit cave halls, highlighting the timeless craftsmanship of ancient Indian master artists.

2. Narrative Storyboard: The Matri-Poshaka Jataka

The centerpiece of the website is an illustrated retelling of the **Matri-Poshaka Jataka** (Jataka No. 455), an ancient Buddhist moral narrative centered on selfless filial devotion, empathy, and integrity. The tale follows the life of the Bodhisattva born as a noble, compassionate white elephant living in the Himalayan wilderness, whose sole devotion is the care and nourishment of his aged, blind parents. The website presents this enduring story across five sequential scenes:

Scene 1 — A Quiet Promise: Secluded in the deep forest away from worldly conflict, the white elephant devotes his physical strength to foraging wild fruits, fresh foliage, and tender lotus stalks for his two blind parents, who rest peacefully beneath an ancient bent Sal tree. His strength is defined not by dominance, but by the quiet patience and responsibility he demonstrates.

Scene 2 — A Stranger on the Path: While roaming the forest ridges, the elephant encounters an exhausted traveller who has lost his path amid dangerous ravines. Acting out of pure compassion and asking for nothing in return, the elephant allows the frightened wanderer to ride upon his back and guides him safely out of the wild forest to the populated highway.

Scene 3 — Trust Betrayed: The kindness freely given by the elephant is met with deceit. Arriving in the royal city of Varanasi, the traveller discovers that the King's state elephant has died and that royal scouts seek a worthy successor. Blinded by greed, the traveller leads royal foresters directly to the elephant's sanctuary. Captured in ceremonial ropes, the elephant offers no violent resistance, looking back with sorrow as he realizes his blind parents are left helpless and waiting.

Scene 4 — A Silent Answer: Brought to the royal palace courtyard, the captive elephant is offered lavish feasts of sugarcane, sweet rice, and fruits. In deep grief for his starving parents, the elephant fasts and refuses all food. Moved by the animal's silent dignity, the King inquires into the cause of his sorrow. Upon learning of the elephant's filial devotion, the King's conscience is deeply stirred.

Scene 5 — The Way Home: Recognizing that true nobility cannot be enslaved, the King grants the elephant unconditional freedom. Returning to the familiar bent Sal tree at dawn, the elephant joyously reunites with his weeping parents with entwined trunks.

Core Moral Lessons: The story highlights universal ethical virtues: filial responsibility (*matri-poshaka*), kindness extended without expectation of reward, the shamefulness of betrayal, and the necessity of empathetic listening in governance.

3. Website Architecture, Design, and Key Features

The website was built following editorial heritage guidelines, emphasizing visual breathing room, restrained chrome, and legible storytelling. Key design and interactive features include:

• **Natural Mineral Pigment Palette:** The visual design faithfully mirrors the organic pigments found in the Ajanta Caves: Basalt Cave Black (#17130F), Raw Umber (#34241C), Indian Red Ochre / *Geru* (#A84F32), Golden Yellow Ochre (#C99A43), Terra-Verte Sage Green (#68765A), and Lime-Plaster Parchment (#E6D2AD), supplemented with a subtle procedural plaster grain overlay.

• **Authentic Photographic Documentation:** Curated, high-resolution photographs of authentic Cave 17 murals and panoramic gorge views provide historical context. Every archival image includes holding collection citations (ASI) and verified open-access licenses (CC0, CC BY 4.0, CC BY 2.0).

• **Five Custom Narrative Vector Illustrations:** To present the Matri-Poshaka narrative truthfully without misattributing ancient murals, five original vector panels were authored for this project. They preserve classical Gupta-Vakataka proportions, tribhanga body contours, lotus-petal eyes, character continuity (the white elephant, blind parents, and bent Sal tree), and period-accurate pavilion architecture.

• **Interactive Story Progress Indicator:** A dynamic reading progress component in the story section continuously tracks the user's reading position across the five scenes using an IntersectionObserver, updating a percentage bar and textual indicator (*SCENE 01 / 05* to *05 / 05*).

• **Scroll-Driven Animations & Performance:** Editorial reveals are orchestrated via GSAP 3 and ScrollTrigger. Animations are non-intrusive, preserving natural scrolling without scroll-trapping. Complete prefers-reduced-motion overrides and no-JS fallbacks are implemented.

• **Responsive Layout & Accessible Navigation:** The interface adapts across devices (320px to 1440px+). Features include a sticky translucent header, an accessible keyboard-navigable mobile drawer, WCAG AA contrast compliance, and skip-to-content accessibility links.

4. Technologies Implemented

Technology	Role and Functional Implementation
HTML5	Semantic page hierarchy (header, nav, main, article, figure, figcaption, footer), descriptive alt text, and ARIA roles.
CSS3	Custom design tokens, fluid clamp() typography, responsive grid and flexbox layouts, media queries, and plaster texture styling.
JavaScript (ES6+)	Modular ES architecture (main.js, navigation.js, story-progress.js, animations.js) managing navigation, drawer toggles, and observers.
GSAP 3 & ScrollTrigger	Scroll-triggered entrance reveals and element timelines, bundled locally in assets/vendor/ for 100% offline functionality.
Git & GitHub	Version control and repository management: github.com/AdityaG346/AJANTA-ECHOES-IN-OCHRE.
Vercel	Production deployment configuration with zero-configuration global CDN delivery. **Live Website:** https://ajanta-echoes-in-ochre.vercel.app/

5. Conclusion

The **Ajanta: Echoes in Ochre** project illustrates how modern front-end web development can serve as a compelling medium for cultural preservation and digital humanities. By synthesizing authentic historical research with modular web standards, custom vector art, and accessible storytelling, the project makes an ancient moral narrative vibrant and relevant for contemporary audiences. Developing this project reinforced core software engineering competencies, including semantic markup, performance budgeting, accessible interface design, and clean separation of concerns, demonstrating that thoughtful web engineering can honor ancient art traditions with fidelity and elegance.